

Exercise 06:

Question-1:

A researcher modifies a dataset to fit a hypothesis that predicts a 20% increase in battery efficiency.

- a) Identify which type of misconduct this represents (Falsification, Fabrication, or Plagiarism).
- b) If the original dataset had 200 measurements and 50 were altered, calculate the percentage of falsified data.

Solution:

a) Misconduct Identification:

The act of modifying an existing dataset to force it to align with a hypothesis (e.g., a predicted 20% increase in battery efficiency) is classified as Falsification. Falsification involves altering, changing, or omitting data to inaccurately represent the research findings.

b) Calculation of Falsified Data Percentage:

Total Measurements (TM) = 200

Altered Measurements (AM) = 50

The percentage of falsified data is calculated as:

Percentage Falsified = (no. of AM / no. of TM) x 100

Percentage Falsified = (50 / 200) x 100 = (1/4) x 100 = 25 %

The percentage of falsified data is 25%.

Question-2:

A researcher submits the same results to two journals to increase publication count.

- a) Define duplicate publication and salami slicing.
- b) Explain why these practices violate publication ethics.

Solution:

a) Definitions:

i) Duplicate Publication:

- 1) Submitting the same research results to more than one journal without proper disclosure. It misrepresents the uniqueness of the study and artificially increases publication numbers.
- 2) Duplicate publication is the act of re-publishing the same research, results, or analysis in more than one journal without informing the editors. It presents previously published work as new, which is misleading and violates the requirement for originality in academic publishing.

ii) Salami Slicing:

- 1) Dividing one complete research work into several smaller papers just to increase the number of publications, even when the pieces do not stand independently.
- 2) Instead of presenting the research as one coherent article, the author spreads fragmented results across multiple papers to increase publication numbers.

b) Ethical Violation:

These practices violate ethics because they skew the scientific record by making findings appear more frequent or robust than they are. They also represent a significant waste of time and resources for editors and peer reviewers who review the same material multiple times. Furthermore, duplicate publication often constitutes a violation of the copyright agreement signed with the first journal.

Question-3:

A journal editor receives a complaint of suspected plagiarism in a submitted manuscript.

- a) Name two international guidelines or organizations that help maintain publication ethics (e.g., COPE, WAME).
- b) Explain the steps the editor should follow according to these guidelines.

Solution:

a) International Guidelines:

i) ICMJE (International Committee of Medical Journal Editors):

This group publishes the Recommendations for the Conduct, Reporting, Editing, and Publication of Scholarly Work in Medical Journals, which sets standards for authorship criteria and conflict of interest disclosure across many scientific disciplines.

ii) OASPA (Open Access Scholarly Publishers Association):

This organization promotes and supports ethical best practices for open access publishers. It collaborates on the Principles of Transparency and Best Practice in Scholarly Publishing, which is a crucial tool for identifying credible journals and avoiding predatory ones.

b) Steps the Editor Should Follow:

i) Verify Allegation:

Immediately suspend the review. Use similarity detection software (e.g., Crossref) to verify the extent and nature of the overlap.

ii) Contact Author:

Confidentially contact the corresponding author, presenting the evidence and requesting a formal explanation for the text similarity.

iii) Final Decision:

Reject the manuscript if deliberate or extensive plagiarism is confirmed. Allow a revision only for minor/negligent citation errors.

iv) Refer & Document:

For severe, confirmed misconduct, the editor should notify the author's institution. All evidence, correspondence, and the final decision must be documented.

Question-4:

A researcher receives an email promising fast publication in a “high-impact journal” for a fee of \$1,500, with no peer review.

- a) Identify why this is considered a predatory journal.
- b) Explain two strategies to verify the credibility of a journal before submission.

Solution:

a) Identification as Predatory:

This is considered a predatory journal because it prioritizes profit over quality, evidenced by the high fee (\$1,500) coupled with the explicit promise of no peer review and aggressive, unsolicited communication. Predatory journals often use deceptive practices like claiming fake impact factors.

b) Strategies to Verify Credibility:

i) Check Indexing:

Verify if the journal is indexed in recognized abstracting and indexing databases like Scopus, Web of Science (WoS), or the Directory of Open Access Journals (DOAJ).

ii) Examine the Editorial Board & Peer Review:

Check if the editorial board members are real, reputable academics, and examine if the journal clearly describes its peer review policy (e.g., double-blind).

Question-5:

A student's thesis report shows 15% similarity index when checked using plagiarism software.

- a) Explain whether this similarity is acceptable and why.
- b) Suggest steps to reduce plagiarism if it exceeds acceptable limits.

Solution:

a) Acceptability of 15% Similarity:

A 15% similarity index is generally considered acceptable for an academic thesis. This is because similarity software flags all matching text, including correctly cited quotations, reference lists, common technical phrases, and institutional names. A score in this range typically indicates proper inclusion of necessary citations and terminology, not extensive misconduct.

b) Steps to reduce plagiarism if it exceeds acceptable limits:

i) Prioritize Paraphrasing:

Re-read the source to understand the idea, then rewrite the concept entirely in your own unique sentence structure and academic voice.

ii) Use Quotes Sparingly:

Use direct quotations only when necessary (e.g., specific definitions). Enclose all direct text in quotation marks and provide the precise citation (including page number).

iii) **Cite Every Source:**

Ensure every idea, fact, and figure borrowed from another source is followed by an in-text citation, even if the text has been paraphrased.

iv) **Check Self-Plagiarism:**

Do not reuse large sections of text from your own previously submitted reports or assignments without proper acknowledgement/citation, particularly in the methodology sections.

v) **Review Technical Text:**

If the software flags common technical or methodical language, verify that the text is truly generic. If not, rephrase or credit the original source where the specific language originated.

Inference:

Maintaining ethical standards in scholarly publishing involves presenting truthful findings, producing original work, and acknowledging all referenced materials accurately. Researchers must avoid falsifying data, duplicating work, or submitting to predatory journals. Following guidelines like COPE and WAME helps ensure integrity and supports reliable scientific progress.